LAB ASSIGNMENT – 6 (Advanced Queries)
DATE – 02.03.2026

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REGISTRATION NO. : AU/2023/0009045

ROLL NO. : UG/02/BTCSECSF/2023/003

SUBJECT : Unstructured Database Lab
SEC: E

Schema

{ emp_id, name, department, designation, city, salary, experience}

Initial Data

Same 10 employee documents as Lab 5.

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Delhi	42000	1

Lab Questions:

Q1. Display employee name and salary only.

```
test> use college_UG_02_BTCSECSF_2023_003
switched to db college_UG_02_BTCSECSF_2023_003
college_UG_02_BTCSECSF_2023_003> db.createCollection("student_UG_02_BTCSECSF_2023_003")
{ ok: 1 }
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.insertMany([
...   { emp_id: 101, name: "Amit", department: "IT", designation: "Developer", city: "Kolkata", salary: 60000, experience: 3 },
...   { emp_id: 102, name: "Neha", department: "HR", designation: "HR Executive", city: "Delhi", salary: 45000, experience: 2 },
...   { emp_id: 103, name: "Ravi", department: "IT", designation: "Tester", city: "Mumbai", salary: 50000, experience: 4 },
...   { emp_id: 104, name: "Pooja", department: "Finance", designation: "Accountant", city: "Chennai", salary: 55000, experience: 5 },
...   { emp_id: 105, name: "Karan", department: "IT", designation: "Developer", city: "Kolkata", salary: 65000, experience: 6 },
...   { emp_id: 106, name: "Sneha", department: "HR", designation: "Manager", city: "Delhi", salary: 70000, experience: 7 },
...   { emp_id: 107, name: "Arjun", department: "Finance", designation: "Analyst", city: "Mumbai", salary: 48000, experience: 2 },
...   { emp_id: 108, name: "Meera", department: "IT", designation: "Developer", city: "Bangalore", salary: 75000, experience: 8 },
...   { emp_id: 109, name: "Rohit", department: "IT", designation: "Team Lead", city: "Pune", salary: 90000, experience: 10 },
...   { emp_id: 110, name: "Anjali", department: "HR", designation: "Recruiter", city: "Kolkata", salary: 42000, experience: 1 }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69c0c8e0945c80a4261e2621'),
    '1': ObjectId('69c0c8e0945c80a4261e2622'),
    '2': ObjectId('69c0c8e0945c80a4261e2623'),
    '3': ObjectId('69c0c8e0945c80a4261e2624'),
    '4': ObjectId('69c0c8e0945c80a4261e2625'),
    '5': ObjectId('69c0c8e0945c80a4261e2626'),
    '6': ObjectId('69c0c8e0945c80a4261e2627'),
    '7': ObjectId('69c0c8e0945c80a4261e2628'),
    '8': ObjectId('69c0c8e0945c80a4261e2629'),
    '9': ObjectId('69c0c8e0945c80a4261e262a')
  }
}
```

Q2. Find employees working in HR department..

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({ ddepartment: "HR" })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q3. Find employees whose salary is between 50,000 and 80,000.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({
...   salary: { $gte: 50000, $lte: 80000 }
... })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  }
]
```

Q4. Display employees from Kolkata city.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({ ccity: "Kolkata" })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q5. Sort employees by experience in ascending order.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find().sort({ experience: 1 })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
```

```
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
```

```
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q6. Display top 5 most experienced employees.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find().sort({ experience: -1 }).limit(5)
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  }
]
```

Q7. Update salary of employee with emp_id 102 to 48,000.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.updateOne(
...   { emp_id: 102 },
...   { $set: { salary: 48000 } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q8. Increase salary of all IT employees by 10,000.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.updateMany(
...   { department: "IT" },
...   { $inc: { salary: 10000 } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

Q9. Update designation to “Senior Developer” for IT employees having experience more than 5 years.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.updateMany(
...   { department: "IT", experience: { $gt: 5 } },
...   { $set: { designation: "Senior Developer" } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Q10. Delete employees whose experience is less than 2 years.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.deleteMany({
...   experience: { $lt: 2 }
... })
{ acknowledged: true, deletedCount: 1 }
```

Q11. Display employees who are not from Delhi.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({
...   city: { $ne: "Delhi" }
... })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```

Q12. Find employees whose designation is Developer and salary greater than 60,000.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({
...   designation: "Developer",
...   salary: { $gt: 60000 }
... })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  }
]
```

Q13. Display employees working in IT or Finance department.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({
...   department: { $in: ["IT", "Finance"] }
... })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```

Q14. Display employees sorted by department and then by salary.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find()
... .sort({ department: 1, salary: 1 })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
```

```
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```

Q15. Count total number of employees in the company.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.countDocuments()
9
```

Q16. Display employees whose salary is not equal to 50,000.

```
college_UG_02_BTCSECSF_2023_003> db.student_UG_02_BTCSECSF_2023_003.find({
...   salary: { $ne: 50000 }
... })
[
  {
    _id: ObjectId('69c0c8e0945c80a4261e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Kolkata',
    salary: 75000,
    experience: 6
  },
```

```
  {
    _id: ObjectId('69c0c8e0945c80a4261e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Bangalore',
    salary: 85000,
    experience: 8
  },
  {
    _id: ObjectId('69c0c8e0945c80a4261e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Senior Developer',
    city: 'Pune',
    salary: 100000,
    experience: 10
  }
]
```